

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

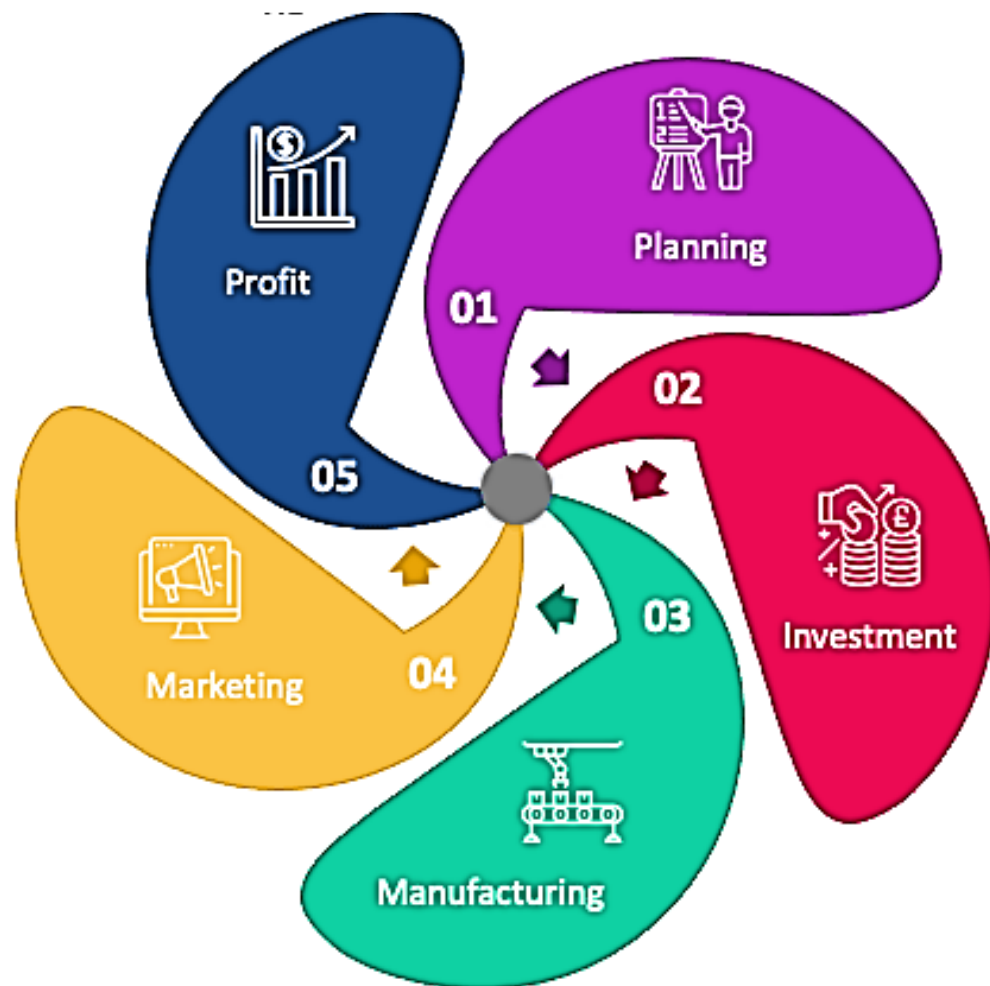
In the name of Allah, Most Gracious,
Most Merciful.



Course HS - 402, Engineering Economics, CR 1, (2+0)/week



Engineering Economics (HS - 407)



Course Instructor

Engr. Yasir Hamid

Lecturer

**Department of Mechanical Engg.
HITEC University, Taxila Cantt**

Contact Details

 **+92-334-5553363**

 **yasir.hamid@hitecuni.edu.pk**

 **yasir.hamid2k09@gmail.com**



Course Outline & Reference Books

Join Google Classroom:

<https://classroom.google.com/c/NzYwNDU5NjkwNDg2?cjc=rk4peiq>

rk4peiq



Course Outline & Reference Books

Reference Books

- 1. Engineering Economy by William G. Sullivan, Elin W. Wicks, C. Patrick Koelling**
- 2. Economics by Campbell R Cornell, Stanley L. Bruce**
- 3. Project Management, the managerial process by Erik W. Larson, Clifford F. Gray**
- 4. Karl E. Case's Principles of Economics (10th Edition)**

Office Visiting Hours

- ❖ **Office 08 MED,**
- ❖ **Phone: Ext 327/329**
- ❖ **Office Hours: 0830 - 1530 hrs.**
- ❖ **Lunch Break 1230 - 1330 hrs.**
- ❖ **Student Hours: 0930 - 1630 hrs. Tuesday**

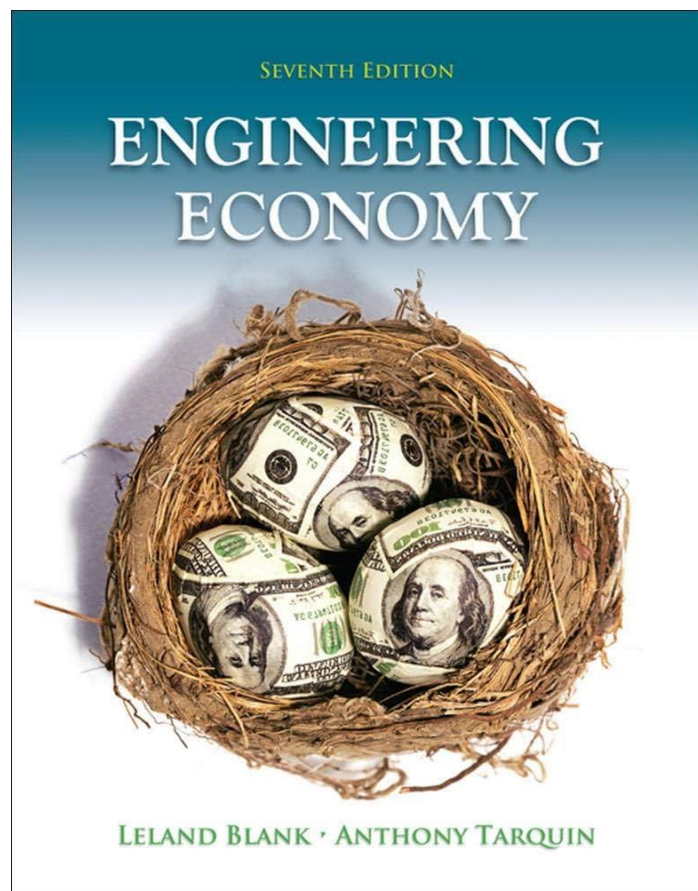


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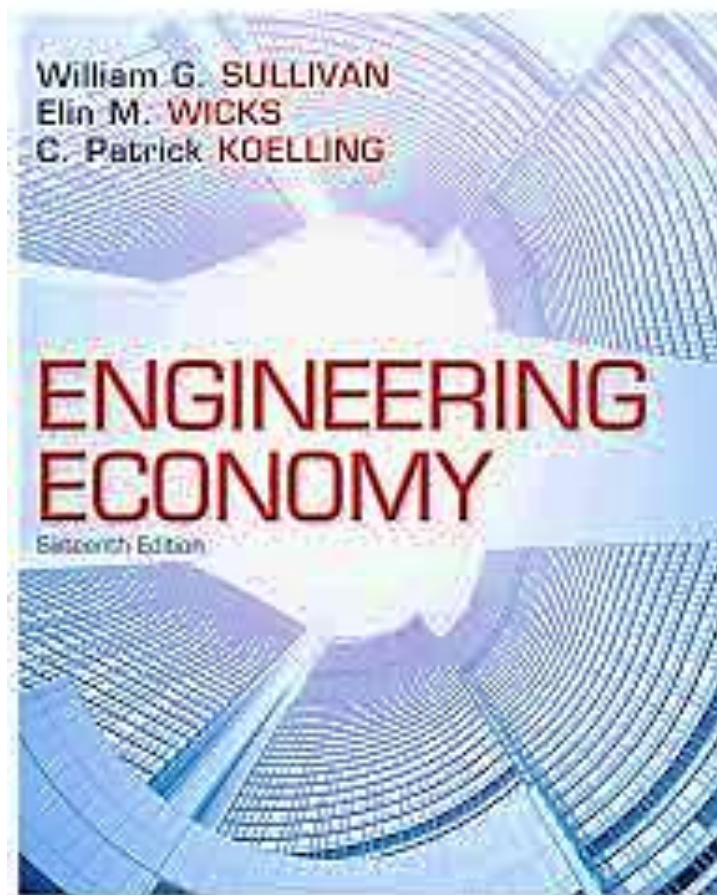


Course Outline & Reference Books

Reference Book 1



Reference Book 2



Reference Book 3





Course HS - 402, Engineering Economics, CR 1, (2+0)/week



Course Outline & Grading Policy

Assignments=10% Quizzes=05% for 1/CH → 2 Quizzes, Project=10% Mid Exam=30% each, Finals=45%

Quiz 1 **Week 03**

Assignment 1 **Week 04**

Quiz 2 **Week 05**

Assignment 2 **Week 06**

Mid Exam **Week 08**

Quiz 3 **Week 10**

Assignment 3 **Week 11**

Quiz 4 **Week 12**

Assignment 4 **Week 13**

Project **Week 14**

Presentation **Week 15**

Finals **Week 17**



Course Outline & Class Policy

- ❖ You are highly encouraged to ask questions and discuss the course material with me and your classmates.
- ❖ No cheating, fabrication, falsification, forgery and computer misuse.
- ❖ Please switch off/put on silent your cell phones.
- ❖ Adhere to the given date for assignment submission.
- ❖ Academic Honesty: Plagiarism will not be tolerated at any level.
- ❖ Extra Help: Do not hesitate to come to my office during office hours or by appointment to discuss a problem or any aspect of the course.
- ❖ University Attendance Policy: Students are expected to attend classes regularly. Minimum attendance must be 75%.



Course HS - 402, Engineering Economics, CR 1, (2+0)/week



Course Outline & Important Note

Make-Up Of Any Exams / Quizzes / Sessional Will Not Be Entertained – Make Sure To Attend It As Scheduled

Make Sure To Comply With The Attendance Policy (75% minimum). No Relaxation Will Be Granted

**Make Habit of Self-Study – Make Habit of Reading Books
Make Use of the Power of Internet for Research**

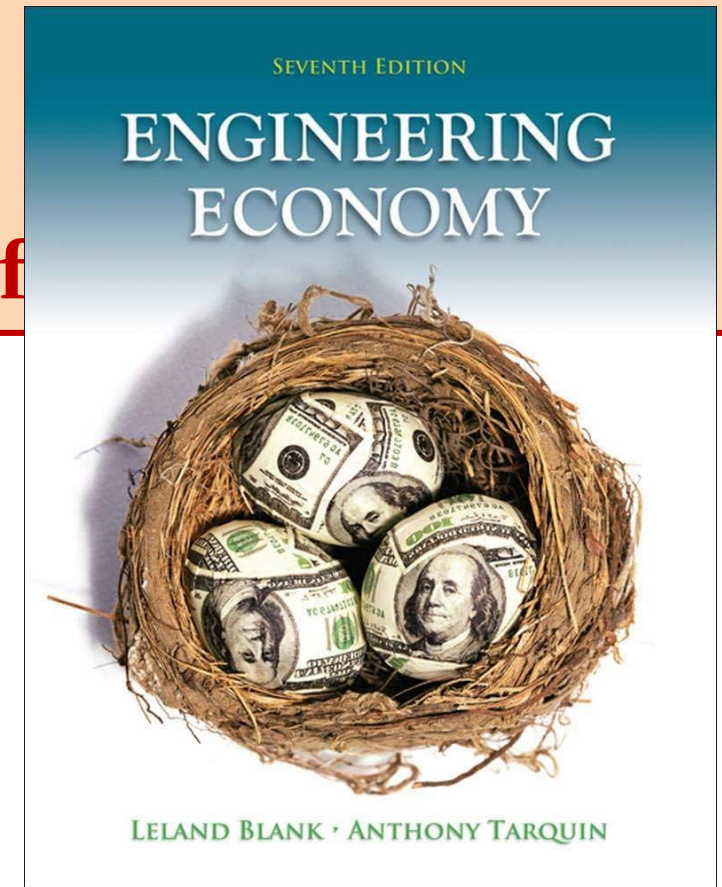


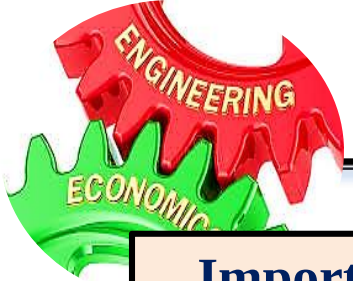
Time Value of Money & Interest Calculations

Chapter # (2)

Factors: How Time and Interest Aff

The Foundation of Engineering Economics by Leland Blank





Important Terminologies

- ✓ Adjusted (cost) basis
- ✓ Unadjusted cost basis
- ✓ Book Value (BV)
- ✓ Market Value (MV)
- ✓ -----
- ✓ Recovery Period
- ✓ Recovery rate
- ✓ Salvage Value
- ✓ Useful life
- ✓ -----
- ✓ Interest
- ✓ Interest Rate
- ✓ Simple Interest
- ✓ Compounds Interest

Simple Interest

Fixed Percentage of the Principal (amount of money borrowed) multiplied by the life of the loan

Simple Interest (SI) = Principal Amount × Interest Rate × Number of Time Periods

If you invest \$1,000 at an interest rate of 5% per year for 3 years, the interest would be:

$$\text{SI} = \$1,000 \times 5\% \times 3 = \$150$$

Compound Interest

Interest that is calculated on both the principal and the interest that has accumulated from previous periods.

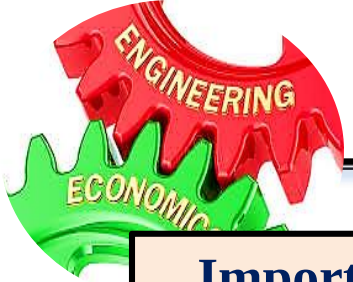
$$\text{Compound Interest (CI)} = \text{Principal Amount} \times \left(1 + \frac{\text{Interest Rate}}{n}\right)^{n \times \text{Time}} - \text{Principal Amount}$$

Principal = The initial amount of money

Rate = The interest rate per period

n = The number of times interest is compounded per year (e.g., annually, quarterly)

Time = The number of years



Important Terminologies

- ✓ Adjusted (cost) basis
- ✓ **Unadjusted cost basis**
- ✓ Book Value (BV)
- ✓ **Market Value (MV)**
- ✓ -----
- ✓ Recovery Period
- ✓ **Recovery rate**
- ✓ Salvage Value
- ✓ **Useful life**
- ✓ -----
- ✓ Interest
- ✓ **Interest Rate**
- ✓ Simple Interest
- ✓ **Compounds Interest**

Compound Interest

Interest that is calculated on both the principal and the interest that has accumulated from previous periods.

$$\text{Compound Interest (CI)} = \text{Principal Amount} \times \left(1 + \frac{\text{Interest Rate}}{n}\right)^{n \times \text{Time}} - \text{Principal Amount}$$

Let's say you invest \$1,000 at an interest rate of 5% per year, compounded annually for 3 years.

$$CI = \$1,000 \times (1 + 5\%)^3 - \$1,000$$

$$CI = \$1,000 \times (1.05)^3 - \$1,000$$

$$CI = \$1,000 \times 1.157625 - \$1,000$$

$$CI = \$1,157.63 - \$1,000 = \$157.63$$

So, after 3 years, you would earn \$157.63 in interest with compound interest, which is slightly more than the \$150 from simple interest.

$$\text{Total amount after 3 years} = \$1,000 + \$157.63 = \$1,157.63$$



Introduction to Engineering Economics (Time Value of Money)

The time value of money (TVM) is a fundamental financial concept that says money available today is worth more than the same amount in the future, because of its potential to earn returns (like interest or investment gains).

If you have \$100 today, you can invest it and earn interest. That same \$100 a year from now isn't as valuable, because you've lost the opportunity to grow it.

Future Value (FV)

$$FV = PV \times (1 + r)^t$$

Present Value (PV)

$$PV = \frac{FV}{(1 + r)^t}$$

Future Value of an Annuity

$$FV = PMT \times \frac{(1 + r)^t - 1}{r}$$

Present Value of an Annuity

$$PV = PMT \times \frac{1 - (1 + r)^{-t}}{r}$$

Present Value (PV): Value of money today.

Future Value (FV): Value of money at a future date.

Interest Rate (r): Growth rate per period.

Time (t): Number of periods (years, months, etc.).

Payments (PMT): Recurring payments, like annuities.



Introduction to Engineering Economics (Time Value of Money)

Emma is planning her finances and is considering two options: (A) She has \$5,000 today and wants to invest it at 6% annual interest, compounded yearly, for 5 years. (B) Separately, she plans to save \$200 every month into a savings account that gives 6% annual interest compounded monthly, also for 5 years. Let's compute each of the following:

Future Value is the amount of money an investment will grow to over a period of time at a given interest rate.
(Compounding)

$$FV = PV \times (1 + r)^t$$

$$FV = 5000 \times (1 + 0.06)^5 = 5000 \times 1.3382255776 = \$6,691.13$$

Present Value tells you how much future money is really worth today if you account for interest. (Discounting)

Emma wants \$10,000 in 5 years, and she can invest at 6%. How much should she invest now?

$$PV = \frac{FV}{(1 + r)^t}$$

$$PV = \frac{10,000}{(1.06)^5} = \frac{10,000}{1.3382255776} = \$7,472.58$$



Introduction to Engineering Economics (Time Value of Money)

Emma is planning her finances and is considering two options: (A) She has \$5,000 today and wants to invest it at 6% annual interest, compounded yearly, for 5 years. (B) Separately, she plans to save \$200 every month into a savings account that gives 6% annual interest compounded monthly, also for 5 years. Let's compute each of the following:

Future Value of an Annuity: Future Value of an Annuity is the value of a series of equal payments made at regular intervals, accumulated at a certain interest rate, at the end of the term. It tells you how much you'll have in the future if you make regular deposits.

$$FV = PMT \times \frac{(1 + r)^t - 1}{r}$$

$$PMT = 200$$

$$\text{Annual rate} = 6\% \Rightarrow r = 0.06/12 = 0.005$$

$$\text{Time} = 5 \text{ years} \Rightarrow t = 5 \times 12 = 60 \text{ months}$$

$$FV = 200 \times \frac{(1 + 0.005)^{60} - 1}{0.005}$$
$$FV = 200 \times \frac{1.348850 - 1}{0.005} = 200 \times 69.77 = \boxed{\$13,954.04}$$



Introduction to Engineering Economics (Time Value of Money)

Emma is planning her finances and is considering two options: (A) She has \$5,000 today and wants to invest it at 6% annual interest, compounded yearly, for 5 years. (B) Separately, she plans to save \$200 every month into a savings account that gives 6% annual interest compounded monthly, also for 5 years. Let's compute each of the following:

Present Value of an Annuity: Present Value of an Annuity is the current value of a stream of future equal payments, discounted at a specific interest rate. It tells you how much a set of future payments is worth today.

Let's say Emma is offered a 5-year contract that pays \$200/month for 5 years, and the interest rate is 6% annually (0.5% monthly).

$$PV = 200 \times \frac{1 - (1 + 0.005)^{-60}}{0.005}$$
$$PV = 200 \times \frac{1 - 0.7416}{0.005} = 200 \times 51.68 = \boxed{\$10,336.82}$$



Introduction to Engineering Economics (Time Value of Money)

Emma is planning her finances and is considering two options: (A) She has \$5,000 today and wants to invest it at 6% annual interest, compounded yearly, for 5 years. (B) Separately, she plans to save \$200 every month into a savings account that gives 6% annual interest compounded monthly, also for 5 years. Let's compute each of the following:

Calculation	Result	What It Means
Future Value (Lump Sum)	\$6,691.13	Her \$5,000 investment grows to \$6,691.13 in 5 years at 6% interest.
Present Value (Future Goal)	\$7,472.58	To have \$10,000 in 5 years, she must invest \$7,472.58 today at 6% interest.
Future Value of Monthly Savings	\$13,954.04	Saving \$200/month at 6% interest gives her \$13,954.04 after 5 years.
Present Value of Monthly Annuity	\$10,336.82	Receiving \$200/month for 5 years is worth \$10,336.82 today (at 6% interest).



Introduction to Engineering Economics (Time Value of Money)

Emma is planning her finances and is considering two options: (A) She has \$5,000 today and wants to invest it at 6% annual interest, compounded yearly, for 5 years. (B) Separately, she plans to save \$200 every month into a savings account that gives 6% annual interest compounded monthly, also for 5 years. Let's compute each of the following:

(Future Value of Monthly Savings)

You decide to put \$200 into a bank every month for 5 years. The bank grows your money at 6% interest.

You want to know: “How much money will be in the bank after 5 years?”

That's the Future Value of Monthly Savings. You're building money over time.

(Present Value of Monthly Annuity)

Now imagine someone promises: "I'll give you \$200 every month for 5 years."

You think: "If I could get all that money upfront today instead of waiting, how much would that be worth?"

That's the Present Value of Monthly Annuity. You're discounting future payments to see their value today.



Time Value of Money and Interest Rate (Accounting Equation)

Double-entry bookkeeping is a fundamental accounting method that ensures the accounting equation ($\text{Assets} = \text{Liabilities} + \text{Equity}$) always remains in balance. This method requires that every financial transaction affects at least two accounts: one account is debited and another is credited. The debits and credits must always balance, meaning the total debits must equal the total credits for each transaction. This system provides a complete record of each transaction, helping to ensure accuracy, detect errors, and maintain financial integrity.

Debit: An entry on the left side of an account that represents an increase in assets or a decrease in liabilities or equity.

Credit: An entry on the right side of an account that represents a decrease in assets or an increase in liabilities or equity.

Accounts: The categories where financial transactions are recorded. Common accounts include:

- **Asset accounts:** Cash, Accounts Receivable, Inventory, Equipment
- **Liability accounts:** Accounts Payable, Loans Payable
- **Equity accounts:** Owner's Equity, Retained Earnings
- **Revenue accounts:** Sales Revenue, Service Income
- **Expense accounts:** Rent Expense, Salaries Expense, Utilities Expense



Time Value of Money and Interest Rate (Accounting Equation)

Equity refers to the ownership value in an asset or business after subtracting any liabilities or debts associated with that asset. In accounting, equity represents the residual interest in the assets of an entity that remains after deducting its liabilities. Essentially, it's what the owners of the business "own" after all debts are paid.

Owner's Equity: This is the capital invested by the owners or shareholders in the business, plus any profits the business has retained over time.

Retained Earnings: The portion of the business's profits that are not distributed to shareholders as dividends but are kept in the company for growth, expansion, or to pay off debt.



Time Value of Money and Interest Rate (Accounting Equation)

Imagine a business has:

- **Assets:** \$150,000 (cash, equipment, inventory, etc.)
- **Liabilities:** \$90,000 (loans, accounts payable)

The equity of the business would be calculated as:

$$\text{Equity} = \text{Assets} - \text{Liabilities}$$

$$\text{Equity} = 150,000 - 90,000 = 60,000$$

In this case, the equity of the business is \$60,000, which represents the value that the owners would receive if the company were to liquidate its assets and pay off all its debts.



Time Value of Money and Interest Rate (Accounting Equation)

The accounting equation is the fundamental equation that keeps together a balance sheet. Indeed, it states that assets always equal liability plus equity. The foundation of accounting is the double-entry system which assumes that a company balance sheet can be broken down in assets, and how they get sources (either through equity/capital or liability/debt).

(The Accounting Equation)


$$\text{Asset} = \text{Liability} + \text{Equity}$$

Assets	=	Liability	+	Owners Equity
Cash		Short-term debt		Capital Stock
Receivables		Payables		Retained Earnings
Inventories		Long-term debt		(income earnings retained in the firm)
Equipment				
Buildings				
Land				

Liability

Obligations contracted by an organization.
Just like the Assets, liabilities can be current and non-current



Equity

Amount of money or resources endowed to an organization



Asset

The assets side of the Balance Sheet is divided in Current Assets (which usually have a life cycle of one accounting year) + Non-Current or Long-Term Fixed Assets (with a life cycle of more than one accounting year)





Time Value of Money and Interest Rate (Accounting Equation)

Let's say you start a business and you invest \$10,000 in cash. At this point, your business has no debt (liabilities), and the investment is all equity.

Accounting Equation:

- Assets: \$10,000 (Cash)
- Liabilities: \$0 (No debt)
- Equity: \$10,000 (Owner's investment)

Investment Example

$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

$$10,000 = 0 + 10,000$$

The equation is balanced, as **Assets = Liabilities + Equity**.



Time Value of Money and Interest Rate (Accounting Equation)

Now, let's say you borrow \$5,000 from a bank. This creates a liability (debt), but the business also receives \$5,000 in cash (an asset). Here's how the equation looks after the loan:

Accounting Equation:

- Assets: \$10,000 (Cash) + \$5,000 (Loan) = \$15,000
- Liabilities: \$5,000 (Loan from bank)
- Equity: \$10,000 (Owner's investment)

Borrowing Example

$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

$$15,000 = 5,000 + 10,000$$

The equation is still balanced: **Assets = Liabilities + Equity.**



Time Value of Money and Interest Rate (Accounting Equation)

Let's say your business makes \$3,000 in sales revenue. The business gets the \$3,000 in cash, so the assets increase, and at the same time, the revenue increases equity.

Accounting Equation:

- **Assets:** \$15,000 (Existing assets) + \$3,000 (Cash from revenue) = \$18,000
- **Liabilities:** \$5,000 (Loan)
- **Equity:** \$10,000 (Owner's investment) + \$3,000 (Revenue) = \$13,000

$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

$$18,000 = 5,000 + 13,000$$

Again, the equation is balanced: **Assets = Liabilities + Equity.**

**Revenue
Example**



Time Value of Money and Interest Rate (Accounting Equation)

Now, let's say you use \$2,000 of the business's cash to pay off part of the loan. This reduces both the cash (asset) and the loan (liability).

Accounting Equation:

- Assets: \$18,000 (Existing assets) – \$2,000 (Cash payment) = \$16,000
- Liabilities: \$5,000 (Loan) – \$2,000 (Payment) = \$3,000
- Equity: \$13,000 (No change in equity)

$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

$$16,000 = 3,000 + 13,000$$

The equation is still balanced: **Assets = Liabilities + Equity.**

**Paying of Debt
Example**



Time Value of Money and Interest Rate (Accounting Equation)

Lastly, let's say the business buys equipment for \$4,000 in cash. The asset (equipment) increases, but cash decreases.

Accounting Equation:

- **Assets:** \$16,000 (Existing assets) – \$4,000 (Cash payment) + \$4,000 (Equipment) = \$16,000
- **Liabilities:** \$3,000 (Loan remains the same)
- **Equity:** \$13,000 (No change in equity)

$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

$$16,000 = 3,000 + 13,000$$

The equation is still balanced: **Assets = Liabilities + Equity.**

**Buying Equipment with
Cash Example**



Time Value of Money and Interest Rate **(Notation and Cash-Flow Diagrams and Tables)**

Cash-Flow Diagrams and Tables, commonly used in engineering economics, finance, and project evaluation

**A Cash-Flow Diagram is a visual representation of inflows (receipts) and outflows (payments) of money over time.
It's typically used to analyze investments, loans, or projects.**

Helps visualize when and how money moves.

Useful for identifying trends, planning budgets, or calculating net present value (NPV).

Time axis: Horizontal line showing time periods (e.g., years, months).

Arrows:

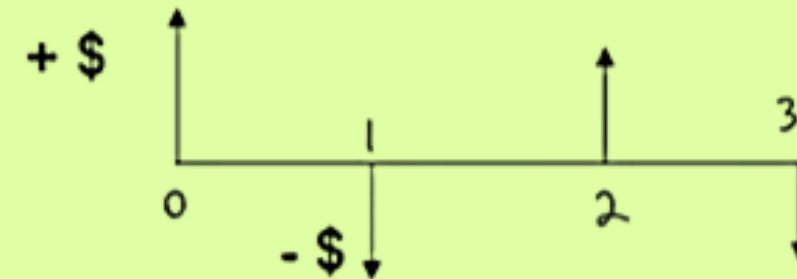
- Upward = Inflows (money received)
- Downward = Outflows (money spent)

Labels: Values are labeled at their respective time points.

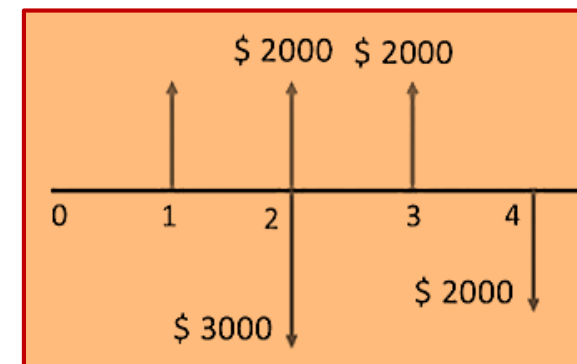


Time Value of Money and Interest Rate (Notation and Cash-Flow Diagrams and Tables)

- Horizontal line divided into period time units
- Vertical arrows represent cash flows; up for benefits/revenues and down for costs/expenses
- Inflows (+) vs Outflows (-)
- Net Cash Flow = Inflows - Outflows



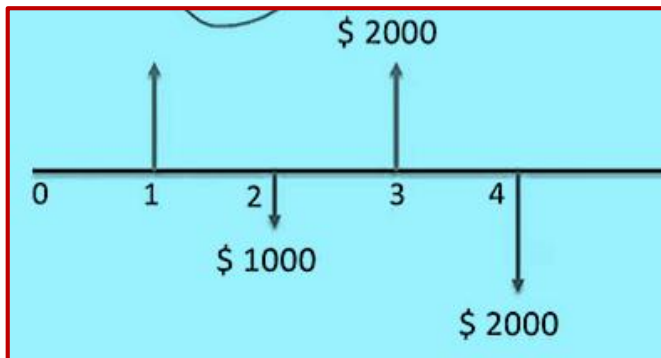
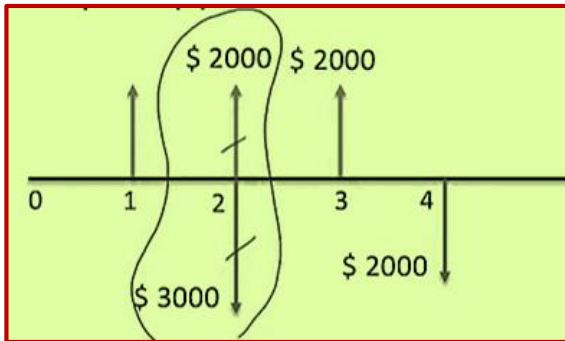
Cash flows cannot be added/subtracted **unless** they occur at the same point(s) in time.



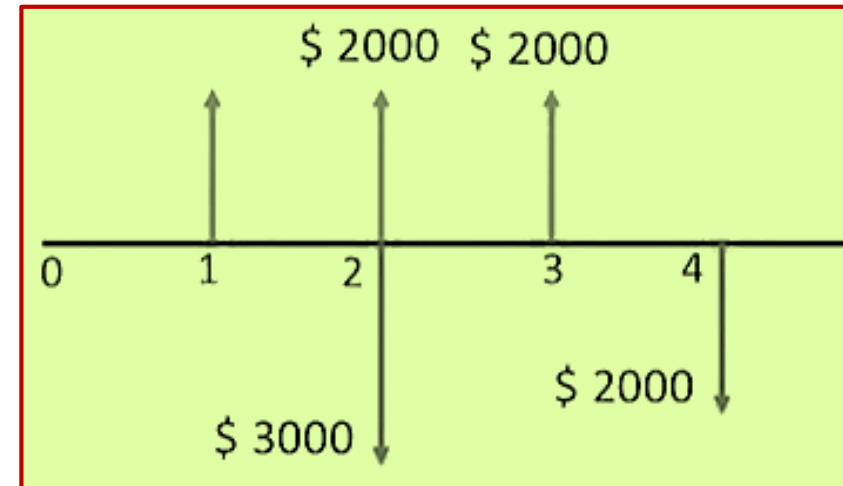


Time Value of Money and Interest Rate (Notation and Cash-Flow Diagrams and Tables)

Cash flows cannot be added/subtracted **unless** they occur at the same point(s) in time.



Cash flows occurring during a period are counted at the **end of the period**





Time Value of Money and Interest Rate (Notation and Cash-Flow Diagrams and Tables)

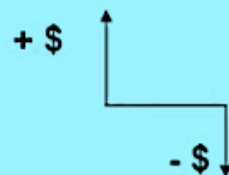
First cost (capital cost, initial investment) = expense to build or to buy and install (-)

Operations and maintenance (O&M) = annual expense, such as electricity, labor and minor repairs (-)

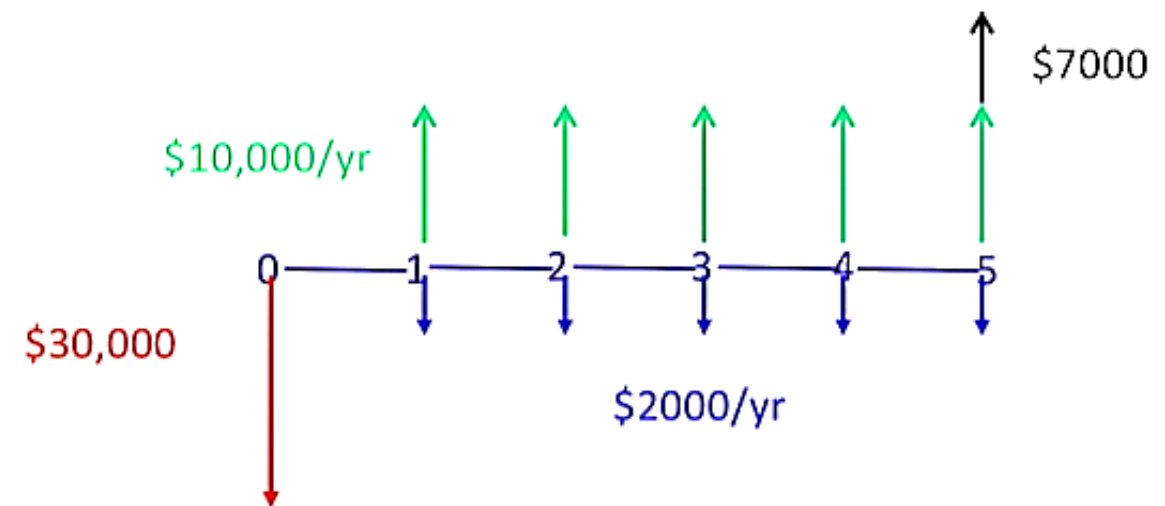
Salvage value = receipt at project termination for sale or transfer of the equipment (+) (can be a salvage cost, e.g., disposal cost (-))

Revenues = annual receipts due from sale of products or services (+)

Overhaul = major capital expenditure that occurs during the asset's life (-)



A machine will cost \$30,000 to purchase. Annual operating and maintenance costs (O&M) will be \$2000. The machine will save \$10,000 per year in labor costs. The salvage value of the machine after 5 years will be \$7000. Draw the cash flow diagram





Time Value of Money and Interest Rate (Example of Cash Flow Diagrams)

A small business is considering installing a solar panel system to reduce electricity costs. Here's what they know:

- Initial investment: \$25,000 (Year 0)
- Maintenance cost: \$500 annually (Years 1–10)
- Electricity savings: \$4,000 annually (Years 1–10)
- Lifespan: 10 years
- Salvage value: \$2,000 at end of Year 10

Step 1: Cash-Flow Table

Year	Description	Inflow (\$)	Outflow (\$)	Net Cash Flow (\$)
0	Initial investment	0	25,000	-25,000
1	Savings – Maintenance	4,000	500	+3,500
2	Savings – Maintenance	4,000	500	+3,500
3	Savings – Maintenance	4,000	500	+3,500
4	Savings – Maintenance	4,000	500	+3,500
5	Savings – Maintenance	4,000	500	+3,500



Time Value of Money and Interest Rate (Example of Cash Flow Diagrams)

A small business is considering installing a solar panel system to reduce electricity costs. Here's what they know:

Step 1: Cash-Flow Table

- Initial investment: \$25,000 (Year 0)
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- Electricity savings: \$4,000 annually (Years 1–10)
- Lifespan: 10 years
- Salvage value: \$2,000 at end of Year 10

Year	Description	Inflow (\$)	Outflow (\$)	Net Cash Flow (\$)
6	Savings – Maintenance	4,000	500	+3,500
7	Savings – Maintenance	4,000	500	+3,500
8	Savings – Maintenance	4,000	500	+3,500
9	Savings – Maintenance	4,000	500	+3,500
10	Savings – Maint. + Salvage	6,000	500	+5,500

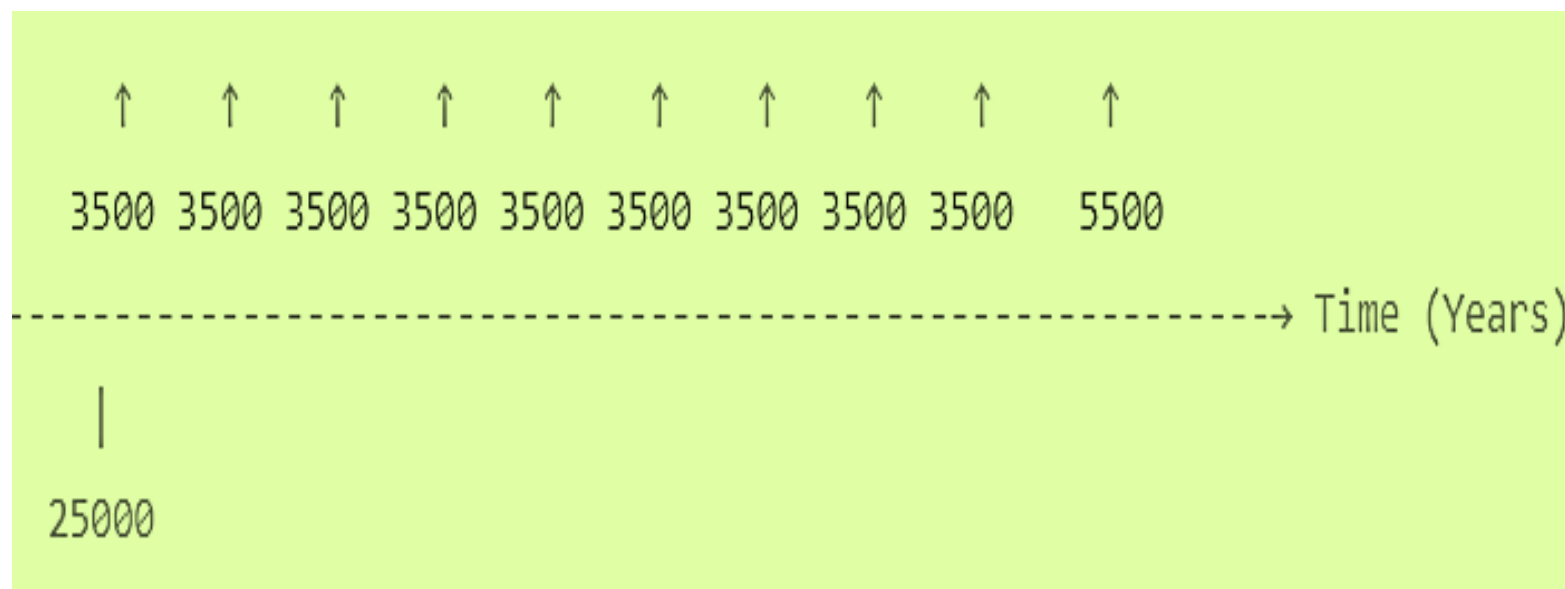


Time Value of Money and Interest Rate (Example of Cash Flow Diagrams)

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- Salvage value: \$2,000 at end of Year 10

Step 2: Cash-Flow Diagram





Time Value of Money and Interest Rate (Example of Cash Flow Diagrams)

Before evaluating the economic merits of a proposed investment, the XYZ Corporation insists that its engineers develop a cash-flow diagram of the proposal. An investment of \$10,000 can be made that will produce uniform annual revenue of \$5,310 for five years and then have a market (recovery) value of \$2,000 at the end of year (EOY) five. Annual expenses will be \$3,000 at the end of each year for operating and maintaining the project. Draw a cash-flow diagram for the five-year life of the project.

As shown in the figure below, the initial investment of \$10,000 and annual expenses of \$3,000 are cash outflows, while annual revenues and the market value are cash inflows. Notice that the beginning of a given year is the end of the preceding year. For example, the beginning of year two is the end of year one.

